Encapsulation

GEF330 – Hiver 2021
CRI03

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Références

- Mathews, Chapter 1.2
- Charles M Kozierok, "TCP/IP Guide – IP Datagram Encapsulation, http://www.tcpipguide.com/free/t_IPDatagramEncapsulation.htm, 2005.

Aperçu

- Encapsulation
- En-têtes des piles de protocoles (Protocol Stack Headers)
- Démo Wireshark

Encapsulation

- Pour garantir que les données se déplacent avec succès à travers différentes couches et technologies, nous utilisons l'encapsulation pour déterminer strictement quelles données appartiennent à quelle couche.
- Considérez-le comme des enveloppes dans des enveloppes.

- Jouons à un petit jeu…

Let’s play a game !

- Je vais choisir deux étudiants.
- Le premier étudiant doit envoyer un document de plusieurs pages au second.
- Il ne peut envoyer qu'une seule page à la fois.
- Il ne peut modifier le document.
- Le destinataire doit pouvoir reconstruire le document complet dans le bon ordre.
- Pour envoyer le message, il ne peut le donner qu'à un autre étudiant juste à côté de lui, qui doit le traiter et le passer à un étudiant à côté de lui et ainsi de suite (interdit de lancer les enveloppes)
- N'importe lequel d'entre vous ne devrait accepter de traiter que les enveloppes qui vous sont spécifiquement adressées.

Next student	User dest / src	Page #	Page

←

Note : par convention les en-têtes à gauche sont envoyés et reçus en premier

Encapsulation – Liaison des données

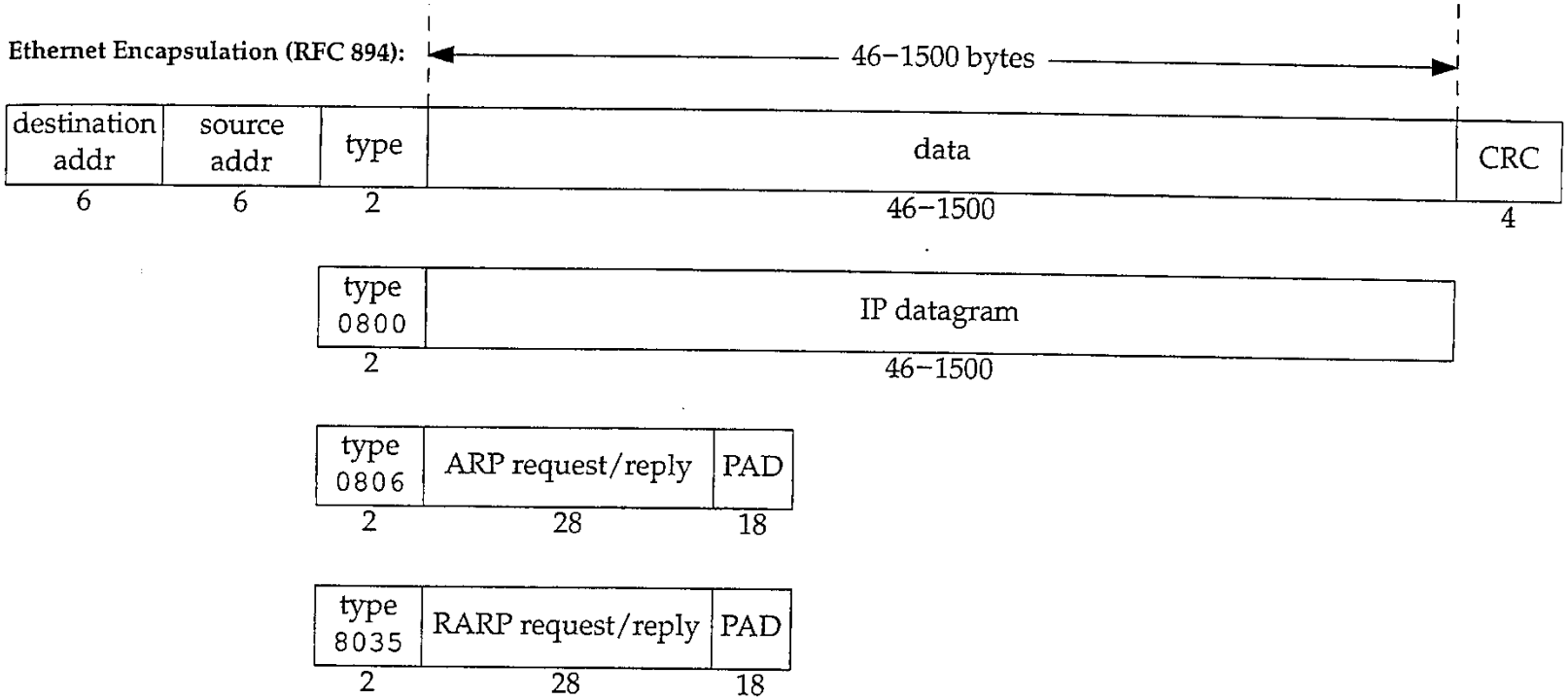
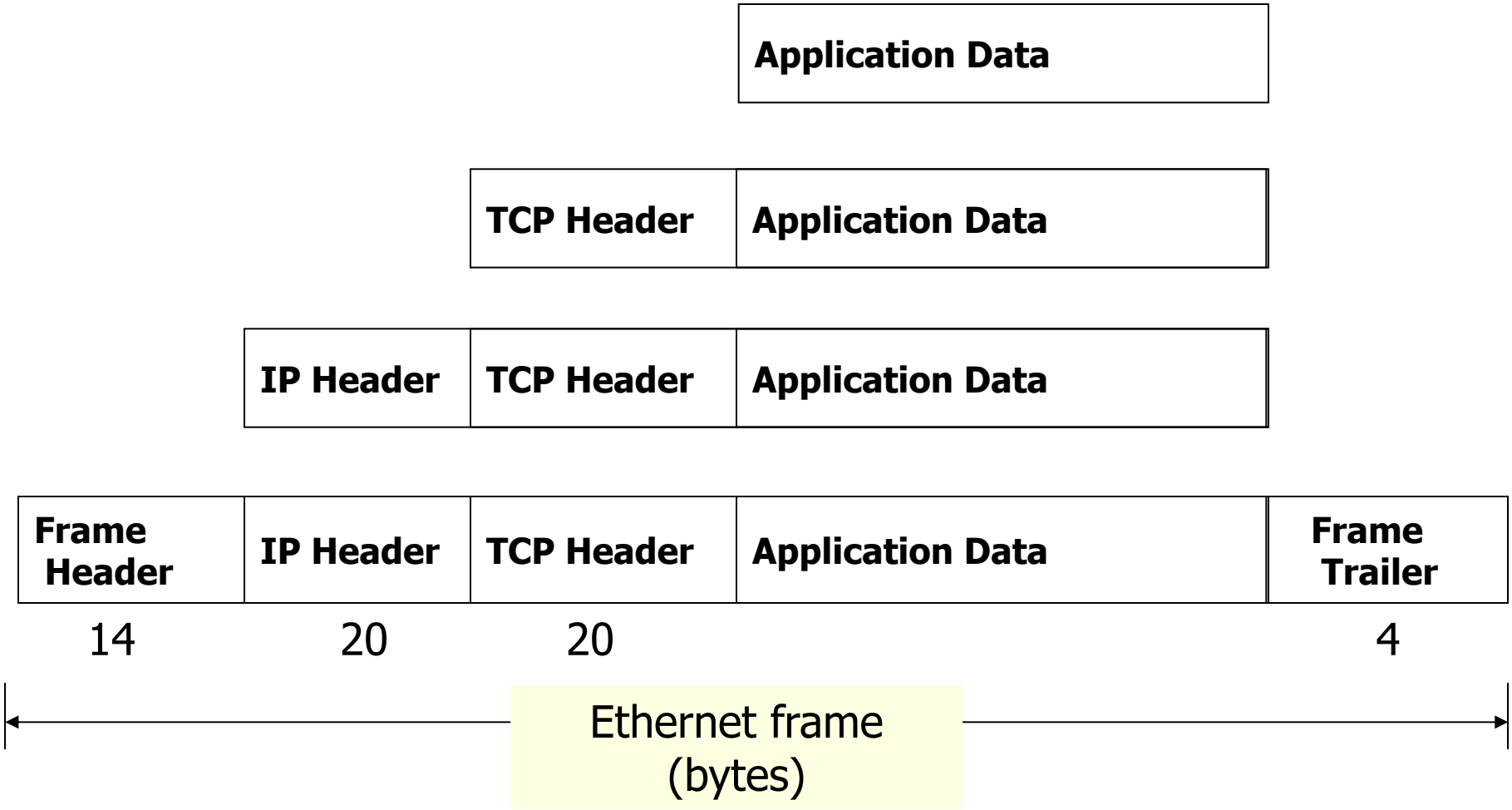
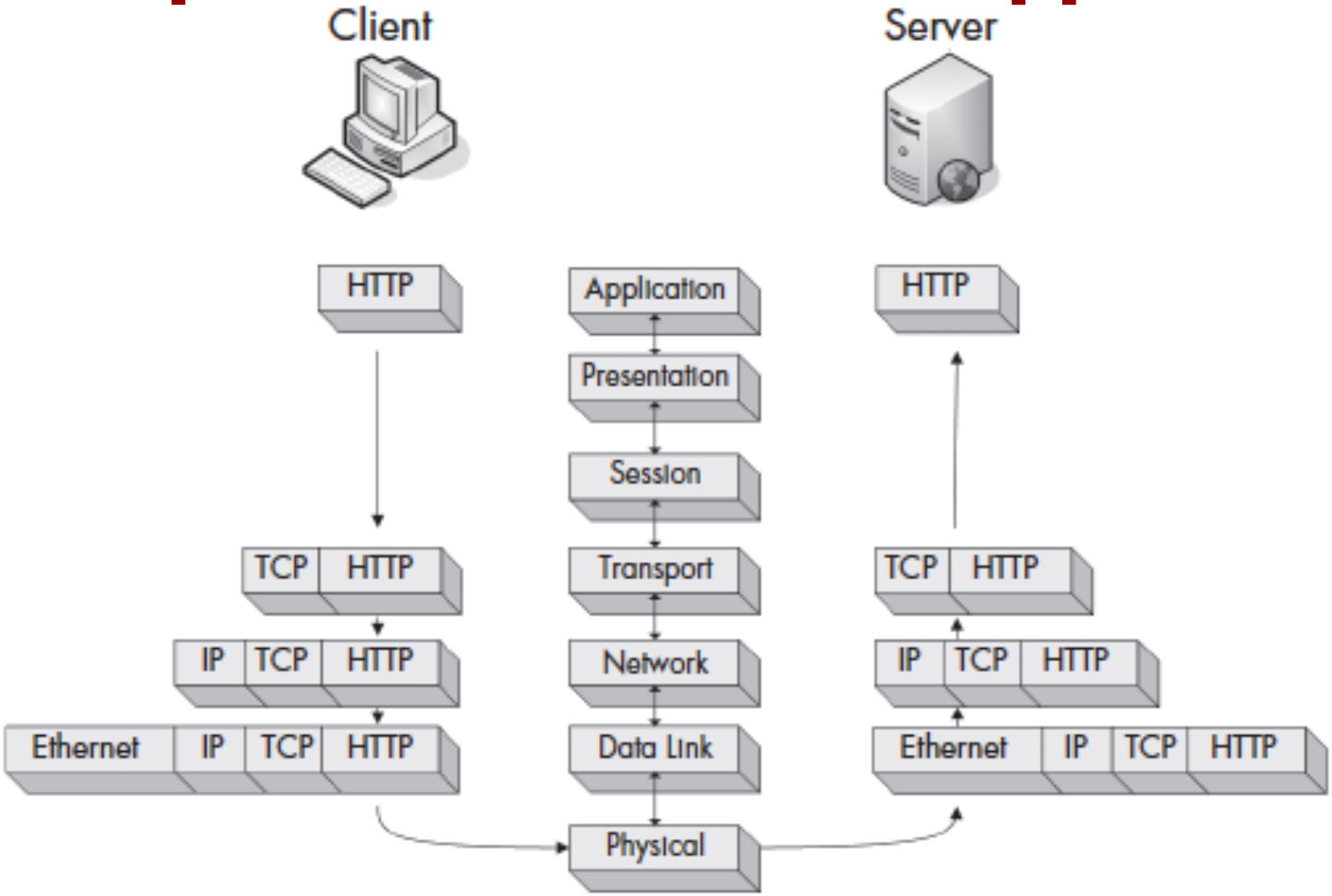


Figure 2.1 IEEE 802.2/802.3 encapsulation (RFC 1042) and Ethernet encapsulation (RFC 894).

Encapsulation – Couche applicative



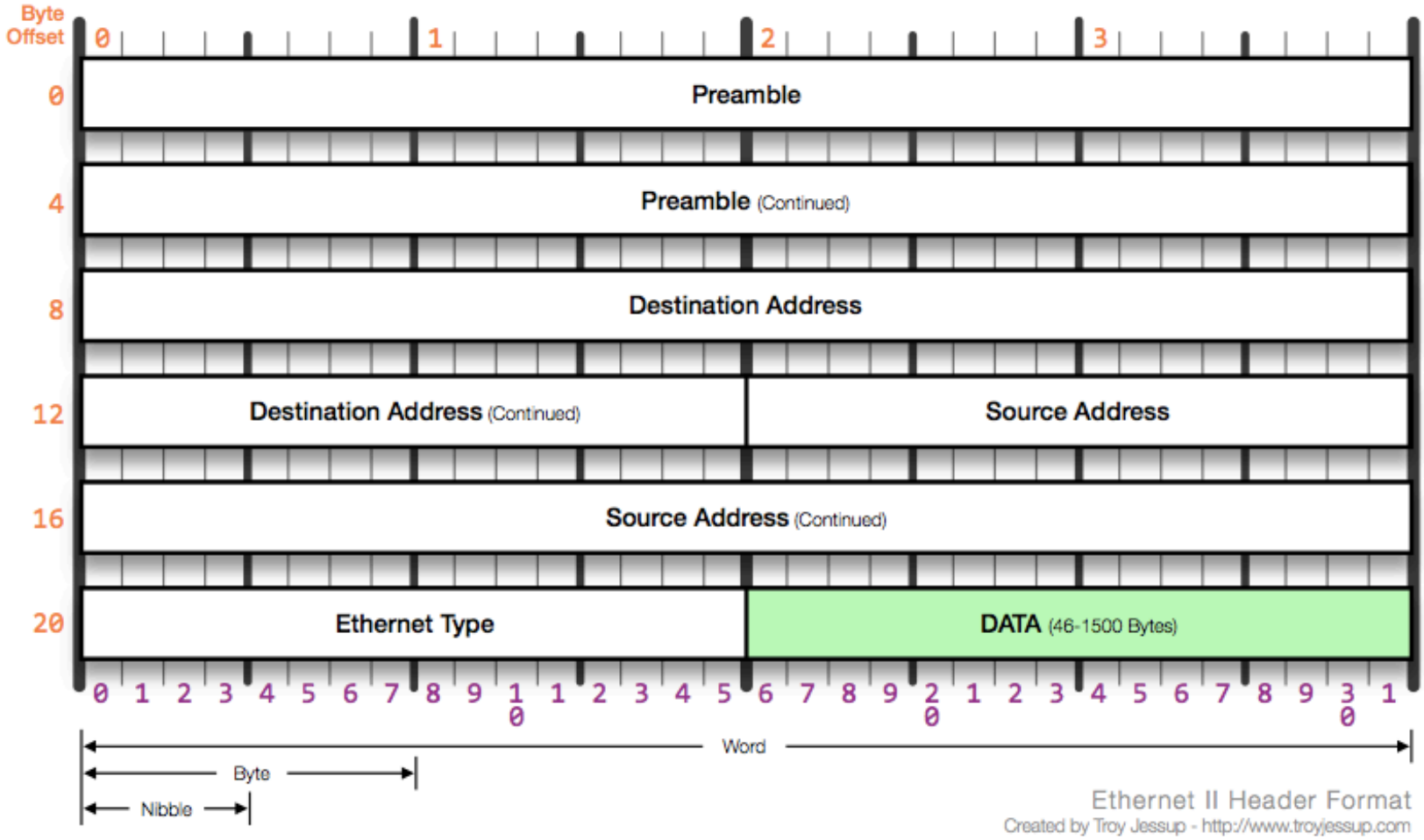
Encapsulation – Couche applicative



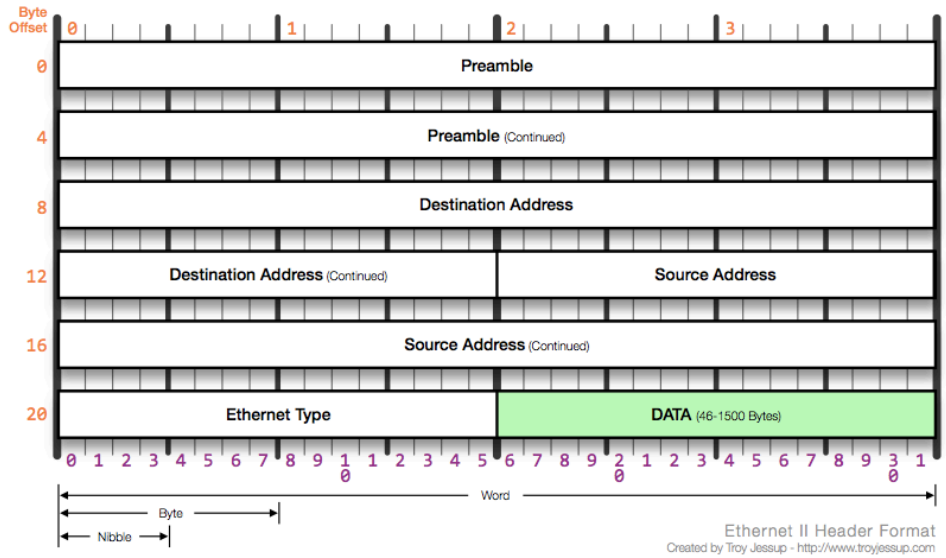
Ethernet Stack Headers

- Les conventions de protocole dictent exactement à quoi l'en-tête doit ressembler.
- Les formats d'en-tête nous indiquent la signification de chaque bit individuel trouvé dans l'en-tête.
- De nombreux en-têtes ont des bits facultatifs, qui peuvent ou non être présents.
  - C'est pourquoi chaque en-tête nous indique soit sa propre longueur, soit l'offset où les données encapsulées peuvent être trouvées.

Ethernet II Header



Ethernet II Header



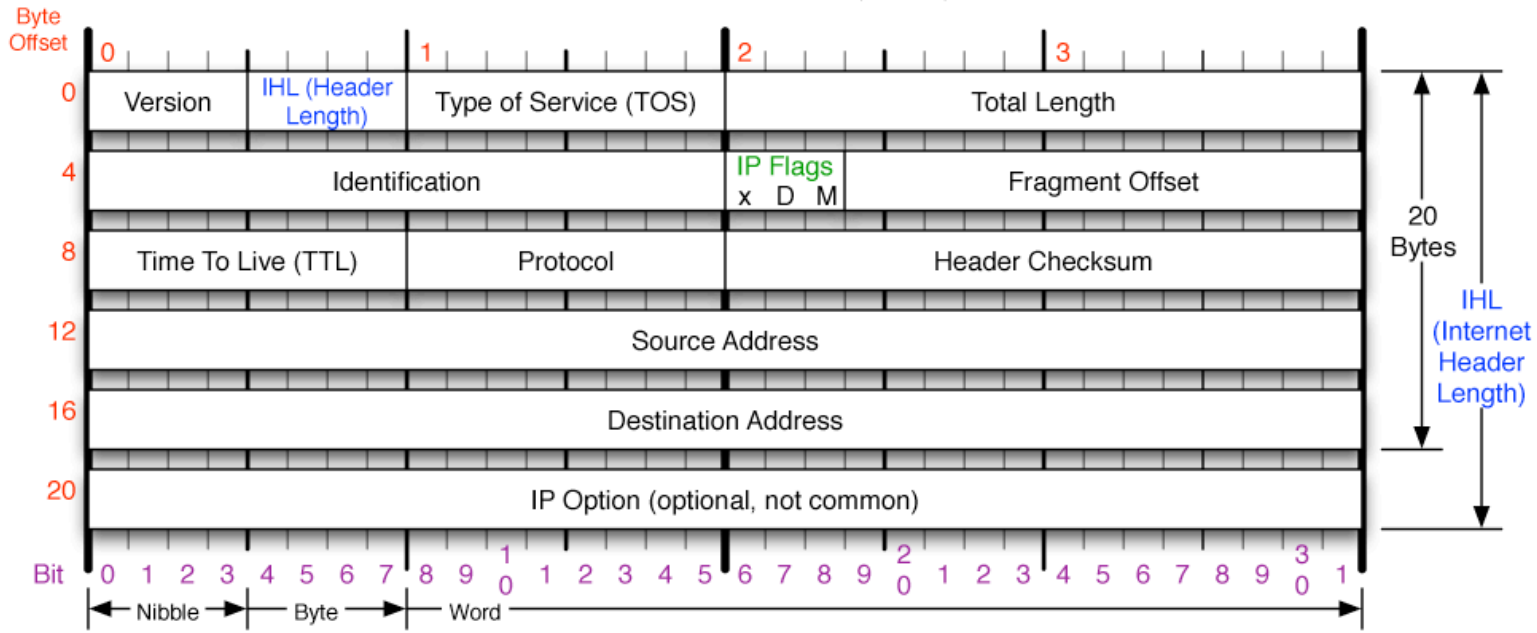
Note: The preamble does not appear in most packet-viewing applications.

Destination Address Source Address Type

```
0006 258d be1d 0007 e953 87d9 0800 4500
0028 12ec 4000 8006 a1ba c0a8 0065 8099
0483 04ca 0016 c6a1 59f9 50dc 3b62 5010
f808 bfe8 0000
```

…and the rest is the encapsulated IP Packet

IP Header (version 4)



Version

Version of IP Protocol. 4 and 6 are valid. This diagram represents version 4 structure only.

Header Length

Number of 32-bit words in TCP header, minimum value of 5. Multiply by 4 to get byte count.

Protocol

IP Protocol ID. Including (but not limited to):

1 ICMP	17 UDP	57 SKIP
2 IGMP	47 GRE	88 EIGRP
6 TCP	50 ESP	89 OSPF
9 IGRP	51 AH	115 L2TP

Total Length

Total length of IP datagram, or IP fragment if fragmented. Measured in Bytes.

Fragment Offset

Fragment offset from start of IP datagram. Measured in 8 byte (2 words, 64 bits) increments. If IP datagram is fragmented, fragment size (Total Length) must be a multiple of 8 bytes.

Header Checksum

Checksum of entire IP header

IP Flags

x D M

x 0x80 reserved (evil bit)
D 0x40 Do Not Fragment
M 0x20 More Fragments follow

RFC 791

Please refer to RFC 791 for the complete Internet Protocol (IP) Specification.

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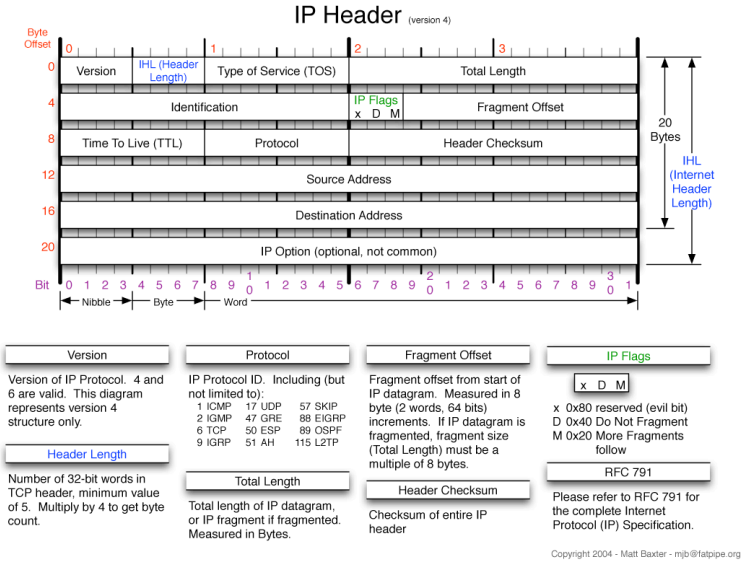
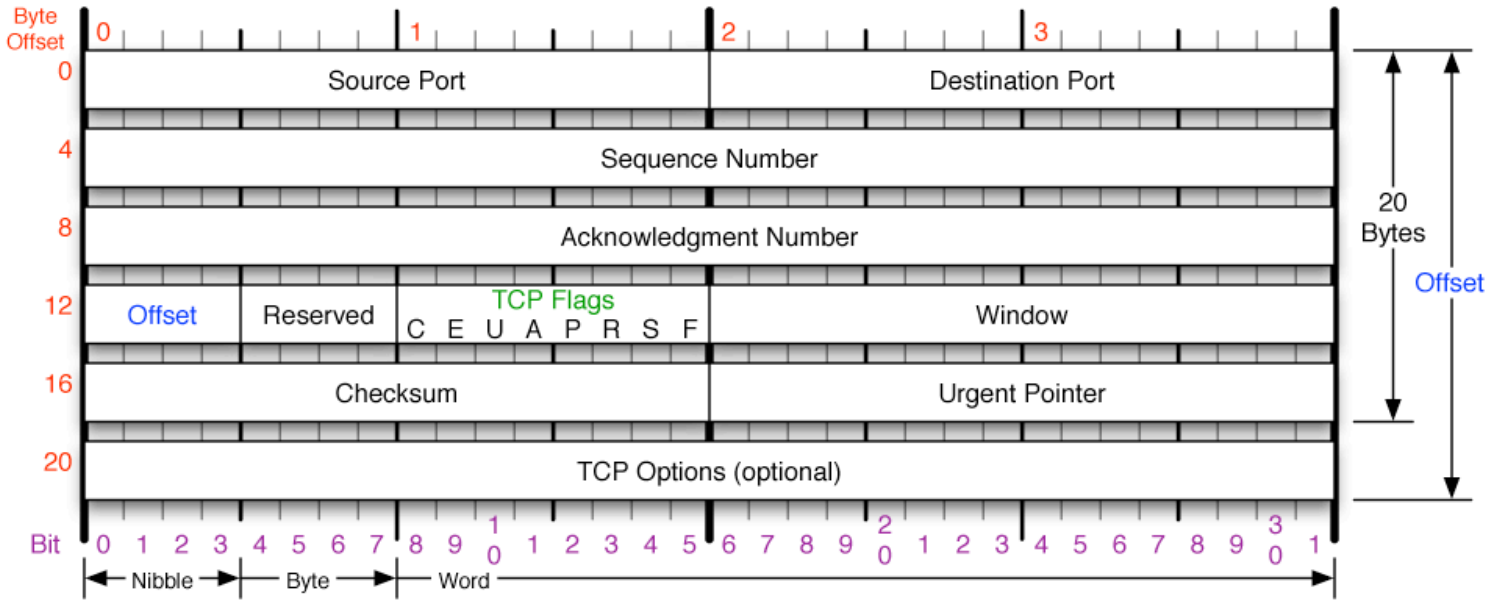


IP Header…

```
4500 0028 12ec 4000 8006 a1ba c0a8 0065
8099 0483 04ca 0016 c6a1 59f9 50dc 3b62
5010 f808 bfe8 0000
```

…encapsulated TCP Data

TCP Header



TCP Flags

C E U A P R S F

Congestion Window

C 0x80 Reduced (CWR)
E 0x40 ECN Echo (ECE)
U 0x20 Urgent
A 0x10 Ack
P 0x08 Push
R 0x04 Reset
S 0x02 Syn
F 0x01 Fin

Congestion Notification

ECN (Explicit Congestion Notification). See RFC 3168 for full details, valid states below.

Packet State	DSB	ECN bits
Syn	0 0	1 1
Syn-Ack	0 0	0 1
Ack	0 1	0 0
No Congestion	0 1	0 0
No Congestion	1 0	0 0
Congestion	1 1	0 0
Reciever Response	1 1	0 1
Sender Response	1 1	1 1

TCP Options

0 End of Options List
1 No Operation (NOP, Pad)
2 Maximum segment size
3 Window Scale
4 Selective ACK ok
8 Timestamp

Checksum

Checksum of entire TCP segment and pseudo header (parts of IP header)

Offset

Number of 32-bit words in TCP header, minimum value of 5. Multiply by 4 to get byte count.

RFC 793

Please refer to RFC 793 for the complete Transmission Control Protocol (TCP) Specification.

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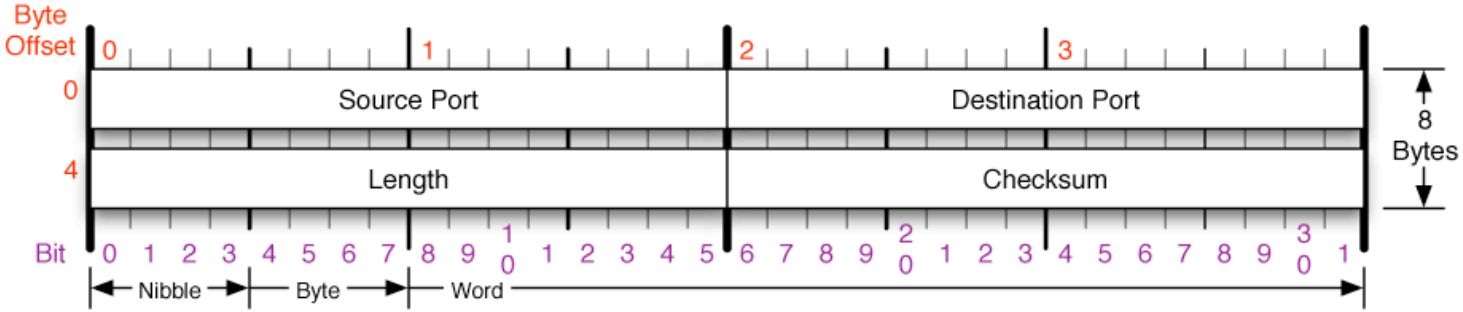
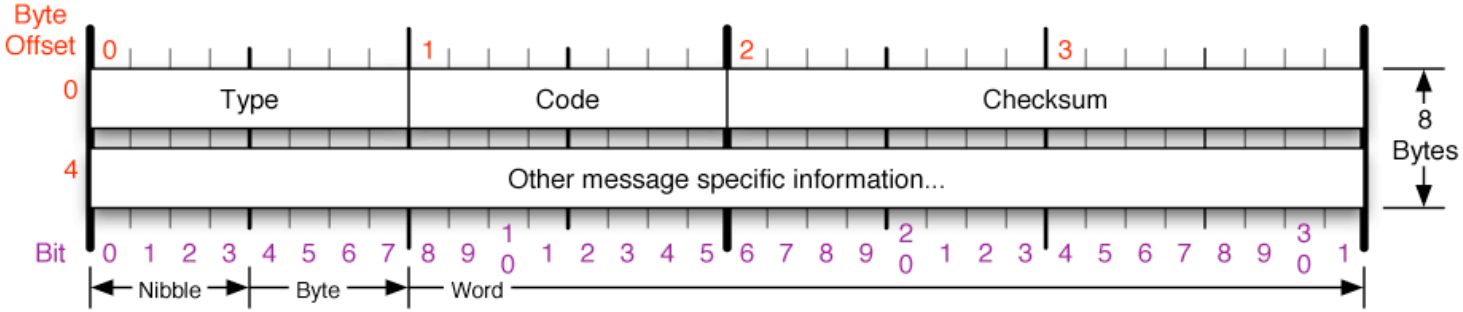
UDP Header



ICMP Header



ICMP Message Types

Type	Code/Name
0	Echo Reply
3	Destination Unreachable
	0 Net Unreachable
	1 Host Unreachable
	2 Protocol Unreachable
	3 Port Unreachable
	4 Fragmentation required, and DF set
	5 Source Route Failed
	6 Destination Network Unknown
	7 Destination Host Unknown
	8 Source Host Isolated
	9 Network Administratively Prohibited
	10 Host Administratively Prohibited
	11 Network Unreachable for TOS
	12 Host Unreachable for TOS
	13 Communication Administratively Prohibited
4	Source Quench
5	Redirect
	0 Redirect Datagram for the Network
	1 Redirect Datagram for the Host
	2 Redirect Datagram for the TOS & Network
	3 Redirect Datagram for the TOS & Host
8	Echo
9	Router Advertisement
10	Router Selection
11	Time Exceeded
	0 TTL Exceeded in Transit
	1 Fragment Reassembly Time Exceeded
12	Parameter Problem
	0 Pointer indicates the error
	1 Missing a Required Option
	2 Bad Length
13	Timestamp
14	Timestamp Reply
15	Information Request
16	Information Reply
17	Address Mask Request
18	Address Mask Reply
30	Traceroute

Checksum

Checksum of entire UDP segment and pseudo header (parts of IP header) (for UDP)

Checksum of ICMP header (for ICMP)

RFC 768 and 792

Please refer to RFC 768 for the complete User Datagram Protocol (UDP) Specification, and to RFC 792 for the Internet Control Message protocol (ICMP) specification.

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Wireshark Demo